

Jeongjin Han

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Research Interests

- Memory-centric systems and data movement optimization
- Large-scale simulation, parallelism, and performance optimization
- System state management, checkpointing, and deterministic execution in large-scale systems

Education

Korea Advanced Institute of Science and Technology (KAIST)

B.S. in Computer Science & Nuclear and Quantum Engineering

Daejeon, Korea

Feb. 2022 – Present

- **GPA:** 3.92 / 4.3 (96.20 / 100)
- **Minor:** Mathematical Sciences
- **Honors:** Dean's List, Spring 2025

Research Experience

VRAIL: Vectorized Reward-based Attribution for Interpretable Learning

Fall 2025

Co-author (equal contribution) | CS377 Reinforcement Learning, KAIST 🔗 [arXiv:2506.16014](#)

- Concretized and implemented a bi-level RL framework (Linear/Quadratic VRAIL) learning interpretable value estimators as potential-based reward shaping functions. Quadratic VRAIL converged **~10% faster** than DQN (538 vs. 600 epochs); Linear VRAIL achieved **100% convergence rate** (10/10 runs) vs. 80% for DQN.
- Led reward transfer experiments showing pretrained VRAIL reward functions generalize to vanilla DQN agents (100% vs. 80% convergence); conducted α -scheduling and environment variant ablation studies; co-authored the manuscript submitted to arXiv.

Nuclear I&C and Autonomous Operation Lab, KAIST

Winter 2025

Undergraduate Researcher | Advisor: Prof. Jonghyun Kim

- Led pipeline redesign to resolve a GPU underutilization bottleneck caused by a CPU-bound simulator; decoupled policy optimization from environment interaction, significantly improving hardware utilization under system resource constraints.
- Reverse-engineered the process memory layout of a closed-source Windows VM simulator to extract and inject runtime state variables, enabling system integration without source access or binary modification.
- Designed and implemented a TCP/Docker-based distributed execution infrastructure bridging a Windows VM and a GPU server, managing cross-system state synchronization and execution control under memory bandwidth and latency constraints.

Reactor Physics and Transmutation Lab, KAIST

Winter 2024

Undergraduate Researcher | Advisor: Prof. Yonghee Kim

- Conducted Monte Carlo neutron transport simulations using **OpenMC**, analyzing neutron moderation behavior under Thermal Scattering Law (TSL) models from the ENDF/B-VIII.1 nuclear data library.
- Investigated computational and memory characteristics of large-scale Monte Carlo simulation workloads, and analyzed scalability behavior under parallel execution.

Projects

KECC: Optimizing C-to-RISC-V Compiler

Fall 2025

CS420 Compiler Design

- Designed and implemented a full C-to-RISC-V compiler in **Rust**: SSA-based IR optimizer (Mem2reg, GVN, CFG simplification, DCE) reducing memory traffic and instruction count, and an Asmgcn backend with struct/array/float/pointer support and phi-node parallel-move SSA destruction.

- Diagnosed and patched 10+ correctness bugs across all phases (non-deterministic GVN hash collisions, erroneous dead GEP elimination, float calling-convention violations, wrong-block phi-argument removal, etc.).
- Evaluated generated code on **gem5 v25.1 MinorCPU** (cycle-accurate in-order microarchitecture) across 60 C benchmarks; achieved performance **comparable to Clang -O1** (geometric-mean cycle ratio 1.009 $\times$), with 37/60 programs outperforming Clang.

🔗 Jeong-jin-Han/CS420-KECC

SHIFT: Sigmoid-Based Heuristic Invertible Fitness-Landscape Transformation

Fall 2025

CS454 AI-Based Software Engineering

- Proposed and implemented a sigmoid-based invertible fitness-landscape transformation for SBST, enabling hill climbing to escape plateaus; applied AST instrumentation, constant folding, and SSA-like variable-shading for Python source transformation with parallel time-budgeted evaluation.

🔗 Jeong-jin-Han/SHIFT-SBST

Nuclear System Code: Overcooling Accident Simulation

Fall 2025

NQE625 Nuclear System Computational Analysis

- Built a 1-D multi-channel nuclear system code from scratch in **MATLAB**: upwind FVM with semi-implicit time integration across coupled subsystems (fluid momentum/enthalpy, point kinetics with delayed neutrons, fuel heat conduction, steam generator primary-secondary coupling).
- Validated numerical solutions against the industry-standard **MARS** code across 10 physical quantities, demonstrating applicability of coupled-system numerical methods at scale.

Skills

Languages C++, Rust, Python, MATLAB, Scala

Tools & Frameworks PyTorch, gem5, Docker, HuggingFace, L^AT_EX, Git, Linux

Relevant Coursework

Systems	Computer Organization (CS.30101) Operating Systems and Lab (CS.30300) System Programming (CS.20300) Introduction to Computer Networks (CS.30401)
CS Theory	Introduction to Algorithms (CS.30000) Data Structure (CS.20006) Compiler Design (CS.40200) Programming Language (CS.30200)
AI / ML	Introduction to Reinforcement Learning (CS.30707) Machine Learning (CS.30706) Introduction to Artificial Intelligence (CS.40700)
Numerical / HPC	Numerical Methods and Computer Simulation (NQE.30011) Computational Analysis in Nuclear System (NQE.60025)
Mathematics	Linear Algebra (MAS.20012) Differential Equations and Applications (MAS.20001) Analysis I / II (MAS.20041/42) Lebesgue Integral Theory (MAS.40041) Probability & Introductory Random Processes (EE.20010)